

Schrödinger's bridge (or dynamic optimal transport) unifies modern generative models

- Of the many possible (unlikely) ways, which one is the most likely? [Lénoard 2014]

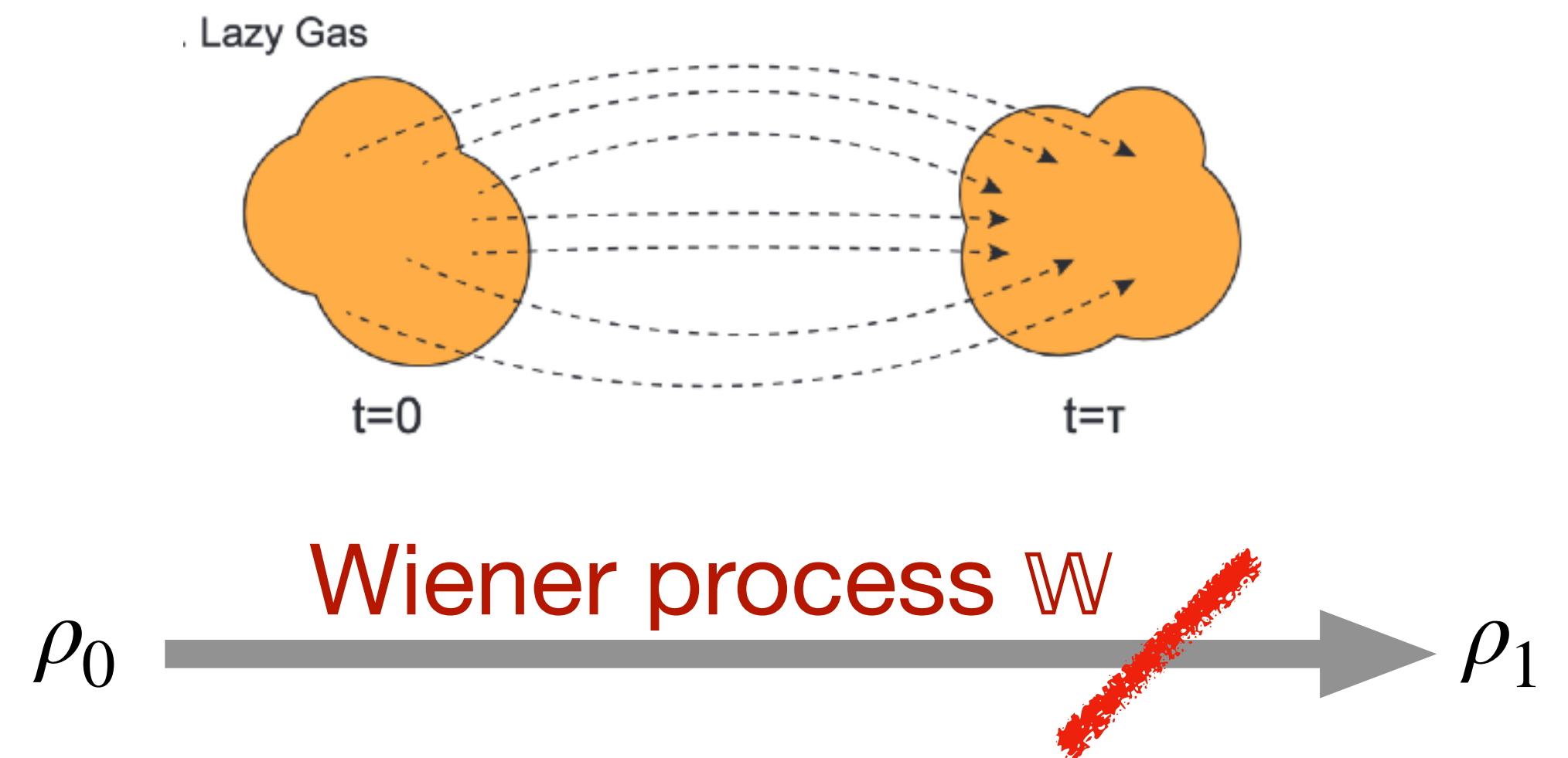
$$\min D_{KL}(\mathbb{P} \parallel \mathbb{W}) \text{ s.t. } \mathbb{P}_0 = \rho_0, \mathbb{P}_1 = \rho_1$$

- A dynamics formulation of entropic-regularized **optimal transport** [Vallani 2009]

$$\min_{\pi \in \Pi(\rho_0, \rho_1)} \int_{\mathbb{R}^n \times \mathbb{R}^n} \frac{1}{2} \|x_0 - x_1\|^2 d\pi(x_0, x_1) + \sigma^2 D_{KL}(\pi \parallel \rho_0 \otimes \rho_1)$$

- A Stochastic optimal control (SOC) formulation

$$\min_{b_t} \mathbb{E} \left[\frac{1}{2} \int_0^1 \|b(t, X_t)\|^2 dt \right], \quad \text{s.t.} \quad \begin{cases} dX_t = [g_t b(t, X_t)] dt + g_t dW_t, \\ X_0 \sim \rho_0, X_1 \sim \rho_1. \end{cases}$$



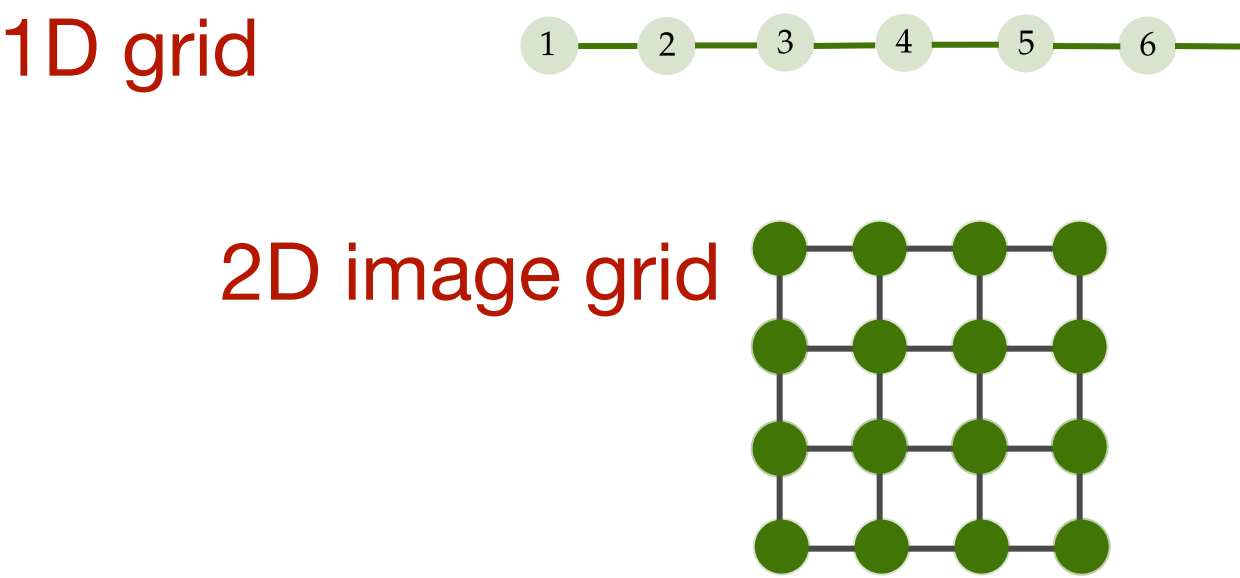
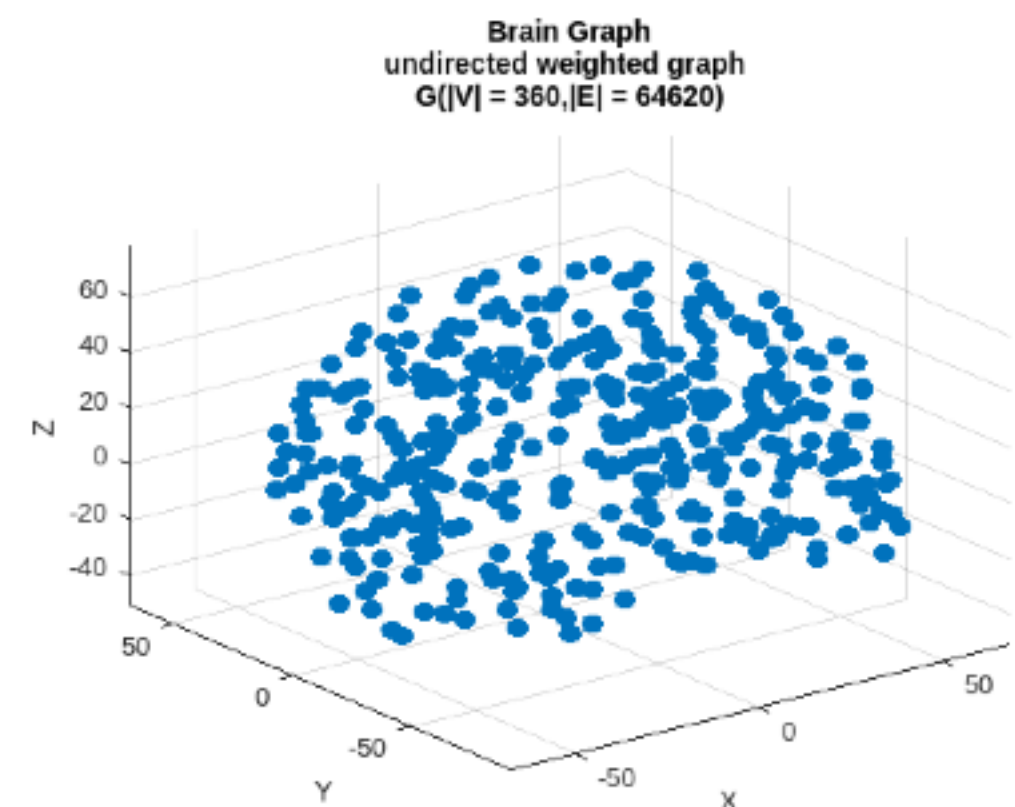
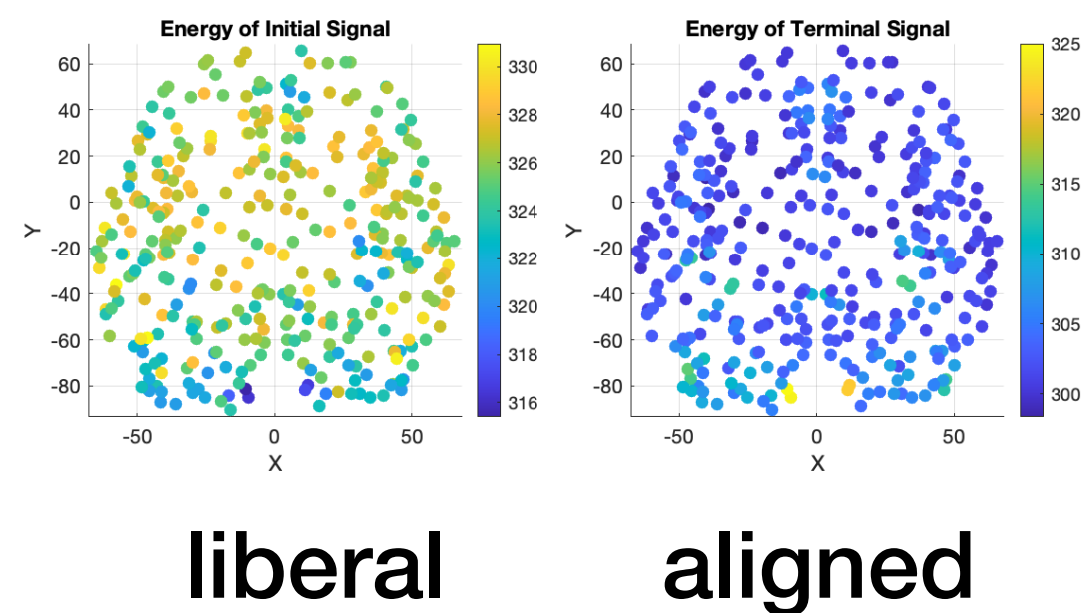
Variational approach

SDE approach

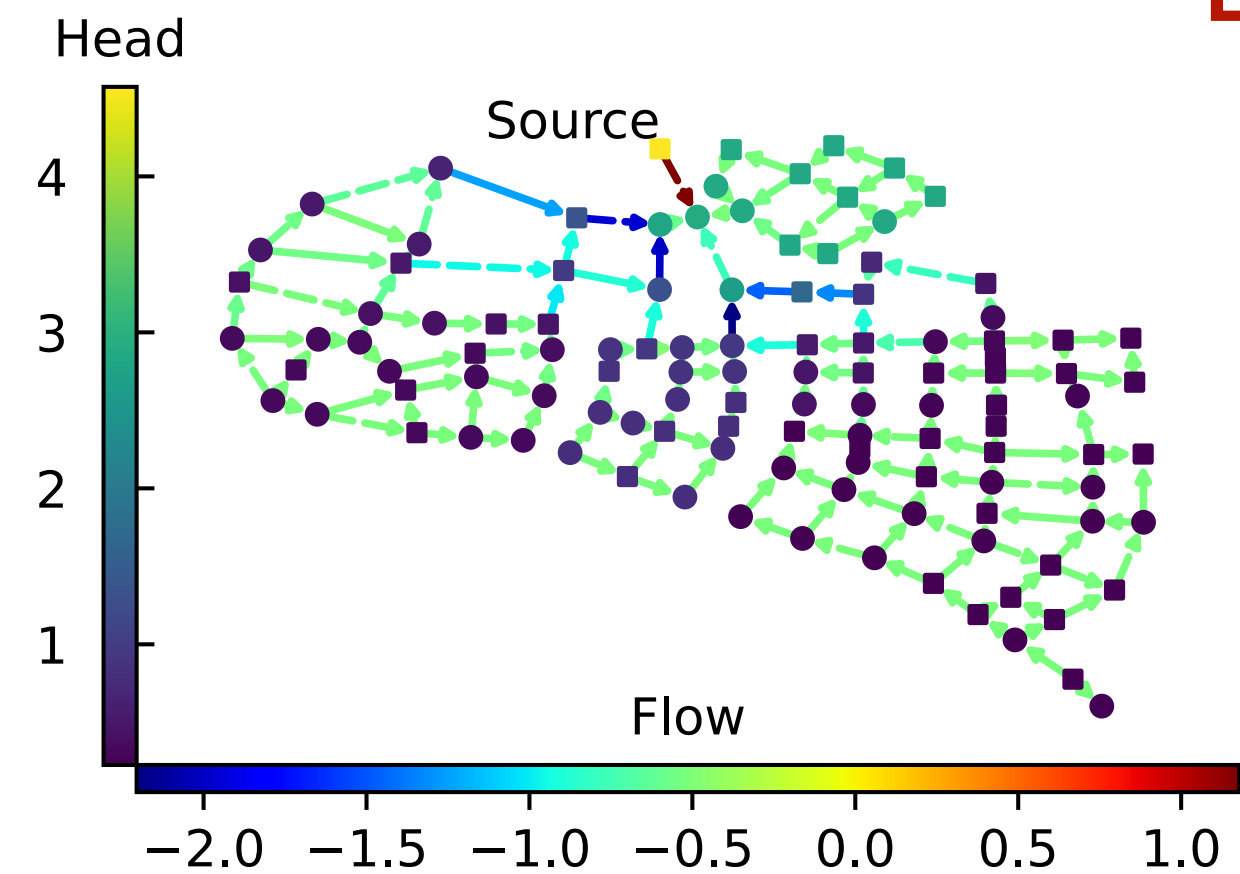
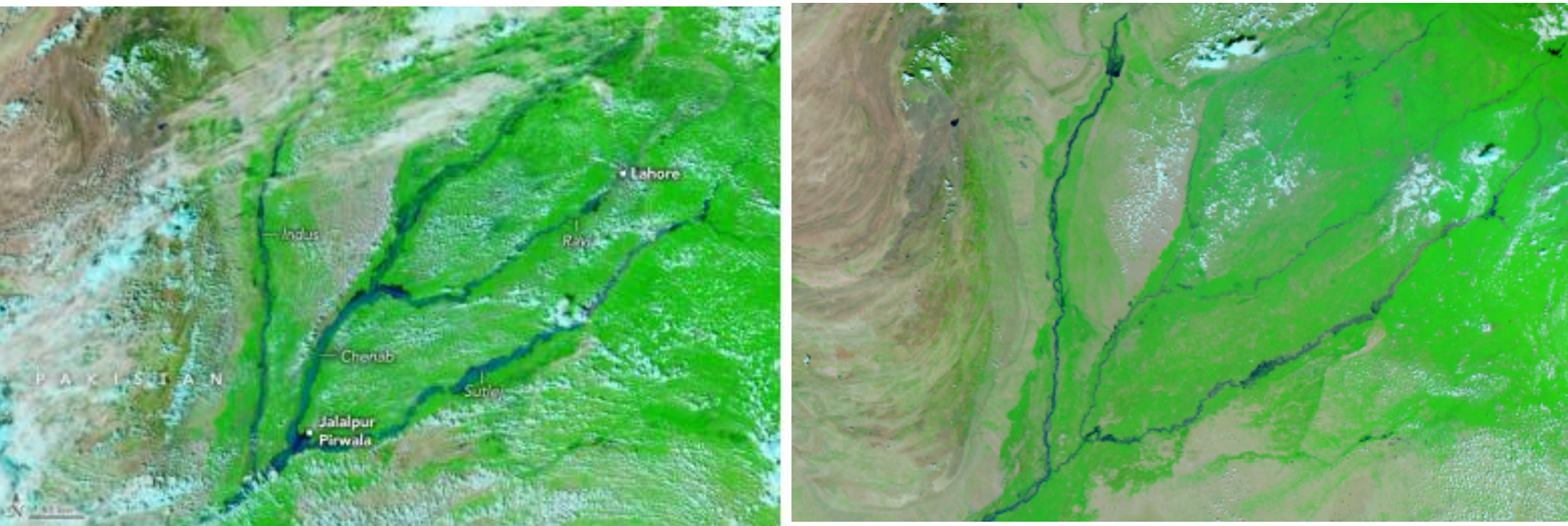
ODE approach

Distribution matching on topology

Brain fMRI signals at different states Dist. of node signals



Punjab river flooding and normal Dist. of edge flows



Existing models ❌